

6(a)(i)	<i>max 2 of:</i> 1 tropomyosin / it, covers / uncovers, myosin binding sites on actin ; R inhibits R active site 2 when calcium ions bind to troponin, tropomyosin / it, moves / changes shape ; 3 allows myosin to, bind to actin / form cross-bridges ; ora	2
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Question	Answer	Marks
6(a)(ii)	<i>max 4 of:</i> 1 ATP hydrolysis / $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$; 2 (causes myosin) <u>head</u> to, pivot / rotate / tilt / stand up ; 3 myosin / head, binds to actin / forms cross-bridges with actin ; R active site 4 ADP and P_i detach ; 5 (myosin) <u>head</u>, swings back / returns to previous position ; 6 actin is moved / power stroke occurs ; 7 (new) ATP binds ; 8 myosin / head, detaches from actin / cross-bridges break ; <i>A mps in any order apart from 1, 4 and 7 which must be linked to correct action</i>	4
6(b)(i)	<i>max 2 of:</i> 1 to, supply / provide, (enough / plenty of) <u>glucose</u> ; 2 for <u>glycolysis</u> ; 3 as little ATP is produced by anaerobic respiration ; 4 as few capillaries are present (to supply glucose directly) ;	2
6(b)(ii)	<i>max 2 of:</i> 1 to, supply / provide, (enough / plenty of) <u>oxygen</u> ; 2 aerobic respiration / oxidative phosphorylation ; 3 to remove, carbon dioxide / lactate ; A lactic acid 4 to, avoid fatigue or promote, stamina / endurance (for exercise / work) ;	2